

Section A - Concept Application

S :- 1

SCENARIO

You are a junior developer at a fintech startup building a digital wallet app. The team is debating between Waterfall and Agile methodology. The project has frequently changing requirements and the client expects to review working features every two weeks.

Question: Which development methodology would you recommend for this project, and why? Identify two specific risks the team would face if they chose the wrong methodology for this context.

Ans:-

I recommend using **Agile methodology** because the project requirements change often and the client wants to see working features every two weeks. Agile allows the team to make changes quickly and get regular feedback from the client.

Risks of choosing the wrong methodology:

1. It will be difficult to make changes if requirements change.

2. The client may not be satisfied because they cannot see regular progress.

S :- 2

SCENARIO

You are collaborating on a college project website with three teammates. Just before a

client demo, one teammate pushed incomplete code directly to the main branch,

breaking the layout for everyone else on the team.

Question: Describe the Git workflow your team should adopt to prevent this from happening again. Name at least two Git commands your team should use as part of this workflow and explain the role of each command.

Ans:-

The team should use a **Git Branch Workflow**.

- Every developer should create a separate branch for their work.
- After completing the work, they should create a Pull Request.
- Another team member should review the code before merging it into the main branch.

Git Commands:

1. `git checkout -b feature-name`
 - Creates and switches to a new branch.
2. `git merge feature-name`
 - Merges the completed branch into the main branch after review.

This workflow prevents broken or incomplete code from reaching the main branch.

S :- 3

SCENARIO

You are building an online internship application form for a college placement portal.

The form must collect: applicant name, email address, phone number, preferred

domain (one of: Web, Data, Mobile, AI), and a resume file upload.

The institution requires

that no field be left empty and that the email field only accepts valid email formats —

without writing any JavaScript.

Question: List the appropriate HTML input types you would use for each of the five fields. Explain

how you would use HTML5 built-in validation attributes to enforce the two constraints mentioned,

and why these attributes alone are not sufficient for production-level validation.

Ans:-

Input Types:

- Name → text
- Email → email
- Phone → tel
- Preferred Domain → select
- Resume → file

Validation:

- Use the required attribute so no field is left empty.
- Use type="email" to allow only valid email formats.

Why is this not enough?

HTML validation can be bypassed by users. The server must also check all data for security and correctness.

S :- 4

SCENARIO

You are designing a news portal that displays articles in a 3-column grid on desktop

screens and in a single-column layout on mobile devices. Your team wants to use

Bootstrap's grid system to achieve this without writing any custom media query CSS.

Question: Explain how Bootstrap's grid class system works to produce this responsive layout. Write the Bootstrap class combination you would apply to the article column div, and explain what each class in the combination controls.

Ans:-

Bootstrap uses a **12-column grid system** to make web pages responsive.

Use this class:

`class="col-12 col-md-4"`

Explanation:

- col-12 → Shows one column on mobile.
- col-md-4 → Shows three equal columns on desktop.

This creates a responsive layout without writing media queries.

S :- 5

SCENARIO

You are writing a C program to record daily rainfall data for a month (30 days). You need to store all 30 readings, calculate the monthly average, and then identify which specific days recorded rainfall above that average.

Question: Explain why an array is the correct data structure for this task compared to using 30 separate variables. Describe the two-step logic your program would follow: first to compute the average, then to identify the above-average days — and explain why both steps cannot be done in a single loop.

Ans:-

An **array** is the best choice because it stores many values of the same type in one variable.

Step 1:

Store all 30 rainfall values and calculate the total.

Average = Total ÷ 30

Step 2:

Loop through the array again and print the days where rainfall is greater than the average.

Both steps cannot be done in one loop because the average is known only after all values are entered.

S :- 6

SCENARIO

You are debugging a C program that uses a pointer to traverse a string entered by the user. The program works correctly when the

user types a name, but crashes immediately when the input field is left empty (the user just presses Enter).

Question: Explain why the program crashes on an empty string input and describe the check your pointer-based solution must perform before dereferencing the pointer. How does traversing a string using a pointer differ from traversing it using array index notation, and which approach makes the empty-input bug easier to catch?

Ans:-

The program crashes because it tries to access an empty string.

Before using the pointer, check:

```
if (*ptr != '\0')
```

This checks whether the string is empty.

Difference:

- Pointer traversal uses pointer arithmetic (ptr++).
- Array traversal uses index numbers (str[i]).

Array indexing is easier for beginners, while pointers are faster and more flexible.